# David Pérez Camacho

## Mobile Tech Lead

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Android · iOS · Flutter · KMP | BLE & connected hardware | Custom Android / AOSP | Embedded firmware | AI orchestration & local LLMs

Mobile engineer and tech lead with **15+ years** across **Android, iOS and Flutter**, specialising in apps that connect to hardware over **BLE**. **Mobile Tech Lead at Stark Future**, where I also lead **AI and agentic workflow** adoption across engineering. Outside work I build with my own **Claude Code orchestration system**, including an open-source app with **100k+ users**. I work across the whole stack, from firmware and custom Android to backend and CI/CD.

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| **15+** years in software | **100k+** open-source app users | **4k+** GitHub stars | **>10,000** bikes running our app |

### AI-ASSISTED ENGINEERING

**Claude Code orchestration**

Built my own **multi-agent orchestration system** on Claude Code, coordinating agents across the development workflow. Built and shipped **SCSKiller** end to end with it.

**Local multi-GPU AI**

Run a **multi-GPU local AI setup** (vLLM, llama.cpp, quantized models) wired into the orchestration, so local models share the work with Claude and power product features.

**Team adoption**

At Stark Future, drove **company-wide adoption** of AI-assisted development (Claude Code, Cursor) and built the **shared skills and automations** teams use daily.

### OPEN SOURCE

**SCSKiller · scskiller.com** github.com/BlueHeisenberg/SCSKiller

Open-source Windows app that precompiles DirectX 12 shaders before you play, so games don't stutter. **4k+ GitHub stars** and **100k+ users**.

- Built entirely with my **Claude Code orchestration system**, from the shader pipeline to the website.
- Telemetry and backends for all those users run on a single small VPS for about **€10 a month**.
- **Community support on Discord** with built-in assistance from my **AI orchestration system**, including its local models.

### EXPERIENCE

**Mobile Tech Lead · Stark Future** CURRENT Dec 2023 – Present

Mobile ecosystem for an electric motocross platform: the bike's dashboard app, a custom Android phone that ships with the bike, and a new rider companion app.

- Grew the mobile team **from 3 to 7** and split it into dashboard and companion app teams; product owner for mobile.
- Architect and maintain the **Stark VARG Dashboard app**, running on **tens of thousands of bikes** worldwide, with a **very high-throughput BLE** connection for live data and telemetry uploaded to the cloud, and riders' watches connected for heart rate. Rebuilt it from scratch in **Jetpack Compose** (Coroutines, Retrofit, Room, Koin) while keeping the legacy app running.
- Built multiple **Flutter** apps that connect to the VARG bike over **BLE** and **Wi-Fi**.
- Building a cross-platform companion app with **Kotlin Multiplatform** and **Compose Multiplatform**, sharing UI and logic between iOS and Android; created an internal **NFC** tool for reading bike VINs.
- Own the **custom phone project** end to end: system-level app, **AOSP** and firmware changes, OTA pipeline and manufacturer coordination.
- Designed secure bike-to-phone **BLE connectivity**, with BLE optimizations alongside the embedded team.
- Contributed to the bike's **embedded firmware** across its multi-chip architecture and microcontroller optimizations.
- **CI/CD** with GitHub Actions (Google Play and AWS internal distribution) and Jenkins for firmware builds; AWS Lambda services for mobile backends.

### Chief Technology Officer · NEN (health-tech)

Dec 2022 – Dec 2023

Designed and built the company's tech stack alone for the first months, then managed one engineer.

- Built **two Flutter apps** for iOS and Android, one for parents and one for kids, on a serverless **Firebase** backend (Firestore, Storage, Auth, Cloud Functions), free during clinical trials and the first release.
- Built a content pipeline with **AI in the loop**: a terminal app to add text, Cloud Functions for translation into 7 languages (**OpenAI API**) and voiceovers (**ElevenLabs**), and a CMS-style website to validate it for production.
- Led technical **GDPR and HIPAA** compliance and set up CI/CD and the company's IT processes.

### CTO & Head of Hardware Development Operations · Smiletronix

Aug 2019 – Dec 2022

Joined as Head of Mobile and moved on to lead embedded and hardware development for the company's device.

- Managed **three staff** and, with freelancers, up to **nine people**, plus external partners across hardware, firmware and apps.
- Designed and wrote the **embedded software**, on a custom **Yocto Linux**, for an in-mouth device with **nine cameras**, which streamed images to the phone over **Wi-Fi**, with **BLE** for setup and control.
- Built the **iOS (Swift, SwiftUI)** and **Android (Kotlin)** apps, CI/CD with Bitrise and Bitbucket Pipelines, a custom **Python** backend, and AI inference on GCP, AWS and Azure.

### Freelance Engineer

Jun 2015 – Present

- Designed and built a portable **BLE electro-muscular stimulator** for therapy: PCB design, **Nordic nRF51** firmware and iOS/Android apps.
- Ported a web app to native Android with BLE 4.1 and real-time **C++ through JNI/NDK**.
- **Discovery Channel** Android app; Node.js backend for an AXA-group app.
- Games: ported **The Crow's Eye** (Unity) to macOS and Linux; optimization and DevOps on **Arima** (Unreal Engine 4, Switch/PC).
- Built Rapit, a QR-code restaurant menu platform.

## EARLIER CAREER

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| **Senior Android Developer**, Future Workshops — Riyadh Airports app | 2017–2018 |
| **Senior Main Android Developer**, Monqi — led the Android team; MQTT over AWS IoT | 2017 |
| **Senior Android Developer**, Digital Origin — led Android; Quebueno.es, PagaMasTarde.com, NFC payment pilot | 2015–2017 |
| **Lead Android Developer**, Toro Development — NFC payments wallet; led two engineers | 2014–2015 |
| **Senior Android Developer**, Altran — AXA apps, Samsung Gear (Tizen) | 2014 |
| **CTO & Chief Android Developer**, Devinet — Allianz app with audio/image recognition | 2013–2014 |
| **Android & Web Developer**, V2MSoft, XAL and others — incl. a Bluetooth tool for Manusa doors | 2010–2013 |

## SKILLS

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| **Mobile** | Kotlin, Jetpack Compose, Kotlin & Compose Multiplatform, Swift / SwiftUI, Flutter / Dart, Coroutines, Ktor, Clean Architecture, MVVM, Koin / Dagger, Room, Retrofit |
| **Hardware & embedded** | BLE, NFC, Wi-Fi device streaming, AOSP and custom firmware, Yocto Linux, OTA, Nordic nRF, microcontrollers, C / C++, JNI / NDK, electronics and PCB design |
| **Cloud & DevOps** | Firebase, AWS (Lambda, IoT), GCP, Azure, GitHub Actions, Jenkins, Bitrise; GDPR / HIPAA compliance |
| **Backend** | Node.js, Python, Kotlin (JHipster), REST, MQTT, Socket.IO, Firestore, MongoDB |
| **AI** | Claude Code, Codex, multi-agent orchestration, Cursor, MCP and skills, AI code review, OpenAI API, self-hosted LLMs (vLLM, llama.cpp) |
| **Languages** | Spanish and Catalan (native), English (C1) |

## EDUCATION

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| **Multimedia Engineering studies** — La Salle, Universitat Ramon Llull, Barcelona | 2008–2014 |
| **Multimedia Studies** — Universitat Oberta de Catalunya (UOC) | 2013–2015 |